

BIOPHARMACEUTICS CLASSIFICATION — SUPPLEMENTARY DISCUSSION

eCTD Section 1.14 (Supplementary) | Amlodipine Besylate Tablets 5 mg / 10 mg

Note: This document is a supplementary biopharmaceutics discussion prepared for academic portfolio purposes. It is NOT the basis of submission for this mock ANDA. The submission basis for this dossier is standard in vivo bioequivalence data, presented in Module 5. This discussion is included to demonstrate applied understanding of the Biopharmaceutics Classification System (BCS) and when a biowaiver pathway could, in principle, be considered for a molecule with these properties.

1. Basis for ANDA Submission

AMLOZA (amlodipine besylate) Tablets, 5 mg and 10 mg, are submitted as a generic duplicate of the Reference Listed Drug (RLD), Norvasc Tablets (NDA 019787, Pfizer Inc.), pursuant to Section 505(j) of the Federal Food, Drug, and Cosmetic Act. The submission is supported by in vivo bioequivalence and comparative dissolution data presented in Module 5.

2. Biopharmaceutics Classification (Background)

Amlodipine is described in the literature as a Biopharmaceutics Classification System (BCS) Class I compound: high aqueous solubility across the physiological pH range (1.2–6.8) and high intestinal permeability. This classification is discussed here for academic completeness and is not used as the regulatory basis for this submission.

3. When a Biowaiver Could Be Considered

FDA Guidance for Industry, Waiver of In Vivo Bioavailability and Bioequivalence Studies for Immediate-Release Solid Oral Dosage Forms Based on a Biopharmaceutics Classification System, outlines criteria under which a biowaiver may be requested for BCS Class I drugs, including rapid and similar dissolution of test and reference products, absence of absorption-modifying excipients, and non-narrow-therapeutic-index status. A hypothetical biowaiver request for AMLOZA would require generation of comparative in vitro dissolution data (f2 similarity) across three biorelevant media — data not generated for this mock exercise.

4. Regulatory Judgment Applied in This Portfolio

For this dossier, the conservative and more broadly defensible path — a standard in vivo bioequivalence study — was selected as the documented submission basis, consistent with how amlodipine ANDAs are typically supported in practice. Recognizing both pathways, and explaining the rationale for choosing one over the other, is the intended demonstration of regulatory judgment in this document.